Asymmetric city: spatial separation of historic and economic city centers

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January 2021

Abstract

In this work, we study whether there are market forces that can separate economic city center from the geographic one. We propose a theoretical model to analyze how the distribution of economic activity and population density are affected by positive externalities from historic buildings. We show the existence of social demand for separation of economic and geographic city centers. Moreover, in the long run, an invisible hand can bring city to the most optimal structure without any land regulation and taxation. We demonstrate that a decrease in transportation costs and a rise of the recognition of the importance of architectural heritage, lead to the shifting of business activity from the city center. In this work, we introduce the concept of a "moving" city, demonstrating that if city centers are separable, market forces can shift the whole agglomeration density. This model potentially can explain the nature of nowadays development of Central Business Districts and the origins of the trend to polycentric cities.

1 Introduction

Architectural heritage seems to be a by-product of cities' past economic development. However, at the same time, economic growth makes adjustments to the cityscape and may distort the accumulation of architectural heritage. So, economic development and architectural preservation can be in conflict in form of "Brusselization", indiscriminate demolition of historical buildings which is presented as an unavoidable sacrifice for economic growth. Usually, these changes caused an outcry among the citizens of the cities. The striking examples of "Brusselization" are urban renewal projects in Saint Petersburg (in 2005-2008) and Stockholm (Redevelopment of Norrmalm).

Central Business District of any modern town usually is the most actively developed and rebuilt area in the city. Unfortunately, this development usually leads to the loss of architectural heritage and the historical appearance of cities.

According to the existing economic literature (e.g., Venables et al., 2020), city agglomeration growth results in increasing demand for space in areas with better access to work and amenities. Thus, in order to sustain economic and urban growth, the city center is often subjected to rebuilding, even when its historic value is socially recognized. So, it seems important to find a compromise between positive externalities from historic preservation and urban renewal for economic growth. We believe that the separation of "economic" and "historic" city centers may solve these issues. Urban development will happen in new areas segregated from historical places. In this work, we would like to consider what economic forces can lead to the shifting of economic activity from the historic center.

Nowadays, cities have become a human's natural habitat, and it is reasonable to explore the ways of efficient city organization. So, a sub-optimal urban organization can lead to significant losses of social welfare (e.g., Venables et al., 2020). In this work, we look at tensions between two centers of gravity in the city, business and historical city areas. Studying the optimality of city structure, one can raise the questions: how they should be located relative to each other and who then "wins the battle" for the city center, firms, or residents? Solving these issues, one can propose social policies to a reorganization of the city to make it more comfortable to everyone.

In this work, we study whether there are market forces that can separate the economic city center from the geographic one. The main trigger of these changes can be the beauty externalities of historic city center or another demanded public place. We study how urban configuration changes in presence of these externalities and what then is the most optimal urban structure. In this work, we propose a theoretical model to study how the distribution of economic activity and population density are affected by positive externalities from historic buildings.

Moreover, it seems that the model may have a methodological value and contribute to the growing literature on theories of the city organization. Most of the papers' results are based on the premise of cities monocentricity. It is usually assumed that the whole production is concentrated in the single point called Central Business District (CBD) which is placed in the geographical center of the city. Even those articles that explore the origins of polycentric cities often use models with exogenously given unique city center (e.g., Berliant et al., 2002). A bunch of works relax this assumption and consider firms allocation problem allowing production to spread over some territory. Current work seems to be one the first that endogenize the economic city center place and consider asymmetric city structure. It is a meaningful step towards studying polycentric city configurations. So, modifications of the proposed model may explain the nature of central business district blurring and offer a mechanism for the genesis of polycentric cities.

Our findings support the idea of the proposed solution of the Brusselization problem. In this work, we show that in most situations there exists a social demand for the separation of "historic" and "economic" city centers. So, social welfare may be increased if the historical center will be outside the business area of the city. Moreover, in the case of free migration, in the long run, an invisible hand can bring city to the most optimal outcome without any land regulation and taxation. The mechanism that lies behind these results is based on the assumption of substitutability of two amenities: better access to the workplace and better access to the historic city center. Separating city centers allows remoteness from the economic center to be compensated by better accessibility to historic areas. Then, an opportunity to choose between "consumption" of two centers increases social welfare.

In this work, we describe the mechanism of possible center-shifting policy. Any shift of historic center from the geographic city center affects firms' location choice and can skew the distribution of firms in the city in the opposite direction. In practice, it can be done by temporal subsidies of the rent outside the city at one side, building necessary infrastructure increasing human incentives to reallocate outside the city. Such intra-city migration can awaken market forces that will further separate economic and historic centers. So, urban development will happen in new areas, shifting population and employment density. It can potentially lead to the shifting of the whole agglomeration in the long run. According to the proposed model, the most optimal city structure is sustainable in the long run without any permanent supplementary policies like city zoning and protection of historical zones which are widely used now.

In this work, we get testable predictions that connect city centers organization with such parameters as commuting costs and population density. We show that a decrease of transportation costs and rise of perception of the architecture, recognition of the importance of architectural heritage, lead to the shifting of business activity from the city center. Moreover, it increases the effectiveness of the center-shifting policy. However, if commuting costs are too low or beauty externalities are extremely high all the predictions change. It becomes more optimal to coincide historic and economic city centers. The intuition behind these results is in line with the logic of Rossi-Hansberg (2004), however, the final predictions differ.

In the following subsection, we discuss the existing literature on an urban organization with a focus on papers that discuss firms' allocation problem. In the second section, we introduce a model and the concept of equilibrium. In the third section, we solve the model defining the locations of agents. In the fourth section, we show the existence of the demand on separation of city centers. In section 5 we study the long-run equilibrium. Section 6 is the extension of the model. In section 7, we consider case when residents locate in the center and are surrounded by firms. The section 8 concludes.

1.1 Literature review

Most of the works on urban economics that study city organization assumes the existence of CBD, where the whole production is concentrated. Pushing the allocation of firms and production issues to the background, economists are usually focused on the distribution of citizens around the city center. The first works that tried to model firm allocation choice were published in the 1980s and were focused on the explanation of the worldwide trend of polycentricity. They consider different determinants of firms' behavior. Fujita & Ogawa (1982) proposed one of the first mathematical models explaining the emergence of urban centers. In their classic model, the main forces that shape the city configuration stamps from externalities in production arising from the proximity of firms to each other. Another classical paper written by Tabuchi (1986), which tried to model city formation, took prices and rents as the main exogenous sources of the changes. However, such a simple model did not account for transaction costs - especially, costs for communication, which all the further works emphasized. Anas & Kim (1996) proposed a city model of urban land considering scale economies in shopping as the main determinant of city configuration. In their work, authors not only model firms' behavior but also endogenize the number and locations of city centers.

The foundational paper, which stated the power and the importance of the firms' allocations and solutions of the urban planner was Henderson & Mitra (1996). The result of the proposed edge city model explained economic urban centers formation and stated that likely not all firms allocate in the center of the city. Following work Zhang & Sasaki (2005) calibrates the modified model with multiple inputs, obtaining symmetric competitive equilibrium. Lucas & Rossi-Hansberg (2002) considered distribution of land within a city proposing one of the most developed urban structure models. In their seminal work, it was shown that for a circular city, businesses can allocate both in the center and in the suburbs. The main driving force for that is the restrictions, imposed by the exogenous forces, households, historical development, and governmental forces.

Another bunch of work put the prior role in city formation on the policies and governmental stimulus. Henderson & Becker (2000) is insightful work that models interactions between employee and employers, emphasizing the importance of consideration of the production side in the economy. However, it also studies the effect of building permits, tax-subsidy policy, and regulation by zoning on city formation. It shows that the restrictions on the land market can lead to the extra-growth of the cities, thus, an improper policy can cause negative externalities. This thought was supported by other researchers, who said that the governmental initiatives can amplify positive effects of laid down externalities, like Rossi-Hansberg (2004) and Rossi-Hansberg et al. (2010). In the current work, we also discuss the role of policies in city formation in the context of the

raised issue.

In this work, we rely on the idea that proximity to beautiful sites rises demand and land rent, which then can wash business out of the city central area. Empirically the effect of beauty externality was detected and estimated by Rossi-Hansberg et al. (2010).

The main paper, which anticipates this work is Berliant et al. (2002). The authors consider the firms' allocation problem proposing a relatively simple framework. In the tradition of the existing literature, authors consider knowledge spillover effects that force firms to locate closer to each other as the main source of the city center formation. They build the linear city model to show what city configurations can emerge in the equilibrium and prove that polycentric city configuration is not sustainable. The basic setup of the current work is similar to one proposed by Berliant et al. (2002).

2 Model

In this Section we develop the model of endogenous city organization, focusing on allocation decisions and the effects of the CBD externalities on land-use patterns. For our analysis, we needed a simple enough framework to model the distribution of firms and residents in the city. A relatively simple model with endogenous location of agents was proposed by Berliand et al (2002). They consider possible city configurations in case of knowledge spillovers effects and then show that polycentric city structure cannot emerge in equilibrium. Generally, we extend the framework proposed in their work modifying a model. In our analysis, we principally consider the most common and reasonable case of monocentric city configuration. At the same time, in contrast to most of the models, we look not only at symmetric allocations.

2.1 Basic Setup

Consider a linear city model with uniformly distributed land. There is a continuum of firms and residents of masses M and N respectively, that can allocate along the linear region from -1 to 1. Assume that $M + N = 2$, $M > 0$ and $N > 0$. Each agent can occupy only a unit density of land and is perfectly mobile within the city. So, each point of the region is associated with residential, business, or mixed land-use. Studying city organization we consider the overall distribution of residential and business land in the city.

If $M < N$, then it can be interpreted as each firm hires several workers. If $M > N$, then it seems that residents can live more compactly and production in the city requires more business land. In the case of $M = N = 1$, setup can

be considered as a matching of workers with available vacancies.

As in most of the previous works, for simplicity, all the land is owned by an absentee landlord who can extract firms' profits and households' incomes through the land rent. In the economy, all firms produce homogeneous composite consumption good using land, capital, and labor. Each household decides where to live and work caring only about maximization of net income spent on consumption.

We introduce two possible sources of externalities in the model: "economic city center", μ, and "historic city center", g. Knowledge spillover effects force firms to locate close to each other and are associated with economic center. Historic center generates beauty externalities that attract households.

Workers meet together to increase their productivity or to be able to specialize in production. More central position in the city relatively to other workers increases the number of social connections and decrease costs of exchange of ideas (or transportation of intermediary goods from other firms). The framework of consideration of this knowledge spillover effect is similar to that used by Berliand et al (2002) and, thus, is analogous to the locational potential function approach in Fujita and Ogawa (1982). For simplicity, we exclude penalty for overall dispersion of firms defining the measure of effectiveness of interactions for firm located at z as

$$Q(z) = 1 - j|z - \mu|, \tag{1}$$

where $\mu = \frac{p+q}{2}$ is the mean location of firm sites. Proximity to this "economic center" implies higher knowledge spillovers effects.

Each firm produces homogeneous composite good hiring labor, L, and capital, K. In our particular formulation, firm production function is

$$Y = A(Q(z)J_L(L))^a K^{1-a}, \tag{2}$$

where $0 < a < 1$, $A > 0$, and $J_L(L) = 0$ for $L < N/M$ and $J_L(L) = 1$ otherwise. We use this production functional form trivializing the effect of labor on production in order to be able to derive analytical solution. Actually, standard Cobb-Douglas production function including labor production factor can be used, however, then it will rise difficulties with determination of workers allocation. In our case, we assume that all firms hire the same amount of labor force and that there is no unemployment, artificially define $L(z) = N/M = \frac{N}{N-2}$.

Each firm faces wages $w(z)$, capital rent r, and land rent $R_f(z)$. So, profit of a representative firm is

$$\pi = AQ(z)^a K^{1-a} - w(z)L - rK - R(z). \tag{3}$$

Under perfect competition each firm has normal profit in equilibrium. Then, bid rent function of a firm is $R_f(z) = \max_{z,K} AQ(z)^a K^{1-a} - w(z)L - rK$. The

first-order conditions with respect to z and K are, respectively,

$$\frac{aY\frac{dQ(z)}{dz}}{Q(z)} = L\frac{dw(z)}{dz} + \frac{dR_f(z)}{dz} \tag{4}$$

and

$$\frac{(1-a)Y}{K} = r. \tag{5}$$

Under perfect competition each firm has normal profit in equilibrium, so, combining (4) and (5) we get

$$Y = \frac{R_f(z) + w(z)L}{a}. \tag{6}$$

Substituting (6) into (4) and solving the resulted differential equation results in

$$R_f(z) = V_0 Q(z) - w(z)L, \tag{7}$$

where V_0 is a constant that should be defined. Manipulating (3), (5), (6) and (7), one can get that $V_0 = aA^{\frac{1}{a}}(\frac{1-a}{r})^{\frac{1-a}{a}}$.

A representative household has preferences described by $\hat{U}(c,h,z,x) = U(c)J_h(h) - e|x-g| - v|z-g|$, where $J_h(h) = 0$ for $h < 1$ and $J_h(h) = 1$ otherwise. $U(c)$ is strictly increasing concave function in consumption, c. There is no utility from leisure and, each household is endowed with one unit of labor and, so, just maximizes its net income. h is the amount of land consumed. For simplicity, land use is modeled in a way to assume constant residential and job densities. x and z are the coordinates of household's apartments and working place respectively. g is a location of "historic city center" and e and v capture the strength of its beauty externalities. Proximity to this "historic center" increase household's welfare. The idea is that if the location is closer to the historic area it is more likely to be more beautiful, and, moreover, it allows to visit this center more frequently. At the same time, agents can also benefit from working near the historic city center. It seems that people spend much time at work, so, one can be more happy if his office will be in the beautiful place with better access to cities amenities (for example, bars which can be visited after the work). Note that another interpretation of g is that it is a location of some public services or a park (bar) that should be periodically visited, then e and v can be considered as commuting costs to this area (from home and from the office).

Residents have expenditures only in the forms of commuting costs and land rent. So, they maximize $I = w(z) - t|x-z| - e|x-g| - v|z-g| - R_c(z)$ by variables x and z.

Overall, in the model each household concerns about relative positions of three locations: apartment (home), working place and historic city center. Agent move between these locations with different commuting costs. t, e, and v are

commuting costs between workplace and home, between home and historic center, and between workplace and historic center respectively. Note that they are not necessarily equal as it seems that people visit their workplaces more often than parks and beauty externalities can be relatively low. Beauty externalities at workplace can be less important than at home. Each agent will be happier if home, office and historic city center will be not far from each other. Beauty externalities and transportation costs were modeled in a very similar way to allow different possible interpretations of the framework.

As agents are perfectly mobile within the city, in order to locational no-arbitrage condition holds, at each location x all households should have the same level of utility, I_0. Then, the bid rent function of a resident is

$$R_c(x) = w(z) - t|x - z| - e|x - g| - v|z - g| - I_0. \tag{8}$$

For the rest of the paper, in order to show the mechanisms of the model more precisely, we will assume that $v = 0$. In Section 6, we relax this assumption showing how it affects our main results.

2.2 Equilibrium

The bid rent function should be continuous within the boundaries of the city in equilibrium, as otherwise some agents will face locational arbitrage possibilities and reallocate to places with sharp discontinuities. Moreover, in the long run equilibrium (if the boundaries of the city are flexible) bid rent function should be also continuous at the ends of the city. Otherwise, some residents will face locational arbitrage possibilities and relocate from one end of the city to another, "shifting" the city. Assume that land rent outside the city is equal to 1 and it is the minimal possible land rent in the economy. Then, in case of fixed city boundaries $R(1) = 1$ or $R(-1) = 1$, while in the long run $R(-1) = R(1) = 1$.

Wage function $w(z)$ should be continuous on the interval [-1, 1] in equilibrium, as otherwise, some residents will face job arbitrage possibilities. Sharp discontinuities in $w(z)$ will attract workers as then sharp change in wage can overweight additional commuting costs. In equilibrium all residents to the left of some $\hat{\mu}$ will work to the left of $\hat{\mu}$, while, all "right" workers should work at the "right" business land. Suppose that one agent that lives to the left of $\hat{\mu}$ decides to work to the right of μ, then he will claim for higher compensation because of commuting costs (than workers that commute from the right), thus, firm will prefer to hire someone else. So, in equilibrium the highest wages will be at $\hat{\mu}$.

Following Berliand et al (2002), we similarly define equilibrium of the model.

A competitive spatial equilibrium is a list of quantities $K(z), Y(z)$, prices $r(z), w(z), R(z)$, and population densities $M(z), N(z)$ for $-1 \leq z \leq 1$ such that the following conditions are satisfied:

1. firms profit is maximized: (5),
2. land rent is maximized: (7), (8),

$$R(z) = \max(R_f(z), R_f(z), 1),$$

$$R(z) = R_f(z) \text{ if } M(z) = 1,$$

$$R(z) = R_c(z) \text{ if } N(z) = 1,$$

$$\min(R(z)) = 1,$$

3. firms have normal profits: (6),
4. land market is cleared: for each $-1 \leq z \leq 1$ $M(z) = 1$ or $N(z) = 1$ and $M(z) + N(z) = 1$,
5. labor market is in equilibrium: $\int_{-1}^{1} L(z)M(z)\,dz = N$ and $L(z) = N/M$,
6. population is balanced: $\int_{-1}^{1} N(z)\,dz = N$ and $\int_{-1}^{1} M(z)\,dz = M$.

The long run competitive spatial job-monocentric equilibrium is defined similarly with an additional condition $R(-1) = R(1) = 1$.

3 City structure

We have to find city organization for a given set of parameters and show under what conditions competitive spatial monocentric equilibrium exists. This work seems to be one of the few that relax symmetric structure assumption. We are especially interested in what effect g will have on city configuration and whether it can shift the economic center μ. Note that μ location is defined endogenously, while g is an exogenous parameter. Without loss of generality, assume that $0 \leq g \leq 1$.

3.1 Urban configuration

Locations which is occupied by firms and households are noted as "business" and "residential" areas respectively. p and q are the right and left outer limits of business area, while P and Q are the outer limits of residential area. Depending on values of parameters of the model, various city configurations are possible.

In this work, we principally consider job-monocentric urban configuration which refers to the case when all firms allocate within the interval $[q, p]$. So, all firms

are grouped together somewhere in the city and are surrounded by residents. It is the most reasonable city configuration for analysis which describes most of the cities in the world. In this case, $-1 < q < p < 1$, $p_h = 1$ and $q_h = -1$. Note that then, $N = 2 - p + q$ and $M = p - q$.

For completeness of the analysis, we consider other possible urban configurations that can occur in the framework of proposed model. We define home-monocentric urban configuration as one that is characterized by the grouped households within the interval $[q_h, p_h]$ ($-1 < q_h < p_h < 1$, $p = 1$ and $q = -1$). Corner solutions when all firms are located on the one side of the city and all residents are located on the other side is noted as fully separated city configuration. In such cases, $q = 1$, $p = q_h$, and $p_h = 1$ (or $q_h = 1$, $p_h = q$, and $p = 1$).

Proposition 1. *A competitive spatial job-monocentric equilibrium exists if knowledge spillover effects are relatively high (compare to historic city center externalities effects),* $j > \frac{2t+(2-N)e}{(2-N)V_0}$.

A competitive spatial home-monocentric equilibrium exists if knowledge spillover effects are relatively low (compare to historic city center externalities effects), $j < \frac{-2t+(2-N)e}{(2-N)V_0}$.

Note that this proposition defines necessary conditions for the existence of monocentric urban configurations and does not guarantees their existence. Depending on values of parameters, instead of monocentric urban configurations we can get fully separated ones. In fact, for any j, depending on other parameters, we can have fully separated city configuration.

In this simple model with linear $Q(z)$ and "externality" term $e|g - x|$, urban configuration type is mainly determined by two parameters, e and j. The result is quite intuitive. Knowledge spillover effects force firms to locate closer to each other and stimulate their demand on land near economic city center. Firms want to be near the geographical city center, as then they can pay lower wages to workers to compensate for their lower commuting costs. At the same time, higher e increases residents' willingness to pay for places near historic city center and stimulate them to live closer to this area. Comparing these two effects, one can predict whose land demand will be higher in the center of the city. If firms' willingness to pay is higher, then the center will be occupied by business area.

At the same time, we can define polycentric urban configurations as cases when firms (or households) are grouped in several clusters and are surrounded by residents (firms). So, an example of polycentric urban configuration will be the situation when all firm locate within the intervals $[q_1, p_1]$ and $[q_2, p_2]$, while residents locate at $[-1, q_1]$, $[p_1, q_2]$ and $[p_2, 1]$. It can be shown that in the proposed

framework it is not possible to get polycentric cases. This conclusion largely is a result of way how knowledge spillover effects are modeled.

Proposition 2. *Polycentric urban configurations cannot emerge in equilibrium.*

Note that in this work we largely talk about monocentricity in a sense that one group of agents is surrounded by another one. It is a common way to discuss it (for example, see Lucas & Rossi-Hansberg (2002) and Berliant et al. (2002)). So, in the urban economic literature, usually discussions on polycentricity refers to the consideration of city structures with multiple clusters of firms. However, it also reasonable to reason about polycentricity in terms of clusters of increasing densities of population.

In our model, we assumed constant residential and business densities. At the same time, we can get peaks of land rent around historic and economic city centers, meaning that demand on space there is the highest. So, if we relax constant population density assumption, because of higher demand we can expect increasing densities around g and μ which can be treated as an example of density-polycentric city configuration. When land rent function has two (or more) local maximums, we can interpret as density-polycentric city configuration. It simply happens, when g belongs to the residential area and $e > t$.

3.2 Historic and economic city centers

For the rest of the paper we will consider only job-monocentric urban configuration (and some fully separated equilibria), when all firms locate within the interval $[q, p]$. So, we assume that $j > \frac{2t+(2-N)e}{(2-N)V_0}$. In Section 7, we will also consider home-monocentric case.

Consider a resident that live at location $x = 1$. The wages in the economy should allow him to work at any workplace which located to the right of $\hat{\mu}$. Then, manipulating (8) one can get

$$w(z) = R_c(1) + t(1 - z) + e(1 - g) + I_0, \text{ if } z \geq \hat{\mu}. \tag{9}$$

Similarly, considering resident at $x = -1$,

$$w(z) = R_c(-1) + t(1 + z) + e(1 + g) + I_0, \text{ if } z < \hat{\mu}. \tag{10}$$

Then, substituting (9) and (10) into (8) yields:

$$R_c(x) = \begin{cases} R_c(1) + t(1 - x) + e(1 - g) - e|g - x|, & \text{if } x \geq \hat{\mu}, \\ R_c(-1) + t(1 + x) + e(1 + g) - e|g - x|, & \text{if } x < \hat{\mu}. \end{cases} \tag{11}$$

Combining (7) with (9) and (10),

$$R_f(z) = \begin{cases} V_0Q(z) - \frac{N}{M}(R_c(1) + t(1-z) + e(1-g) + I_0), & \text{if } z \geq \hat{\mu}, \\ V_0Q(z) - \frac{N}{M}(R_c(-1) + t(1+z) + e(1+g) + I_0), & \text{if } z < \hat{\mu}. \end{cases} \tag{12}$$

As wage function should be continuous, one should find the point of intersection of its two branches. At this point wages attains its maximum value, so, $\hat{\mu} = -\frac{2eg+R_c(-1)-R_c(1)}{2t}$. Then, remind that labor demand does not depend on location z. Moreover, all "left" firms hire "left" workers and all "right" firms hire "right" workers. So, $\frac{1+q}{\hat{\mu}-q} = \frac{1-p}{p-\hat{\mu}} = \frac{N}{M}$.

Then, $p = \frac{-N(2t+2eg+R_c(-1)-R_c(1))+4t}{4t}$ and $q = \frac{-N(-2t+2eg+R_c(-1)-R_c(1))-4t}{4t}$. So, economic center of the city locates at

$$\mu = -\frac{N(2eg + R_c(-1) - R_c(1))}{4t}. \tag{13}$$

Remind that land rent function should be continuous within the city, so, $R_c(p) = R_f(p)$ and $R_c(q) = R_f(q)$. Combining (11) with (12) at locations p and q, and noticing that $Q(p) = Q(q)$, one will be able to find I_0 and get

$$R_c(-1) - R_c(1) = (|g-q| - |g-p| - 2g)e, \tag{14}$$

$$I_0 = \frac{e|g-p| + V_0Q(p) + (N/M+1)(e(g-1) + (p-1)t - R_c(1)}{N/M}. \tag{15}$$

So, $R_c(-1) \geq R_c(1)$ if $g \leq \frac{p-q}{2} = M/2$. If $g \leq M/2$, then $R_c(1) = 1$ and $R_c(-1) = 1 + (|g-q| - |g-p| - 2g)e$. Similarly, if $g > M/2$, then $R_c(-1) = 1$ and $R_c(1) = 1 - (|g-q| - |g-p| - 2g)e$. We have to consider three cases of relative positions of the business area and g. Working with (14) we can find boundaries of firms distribution.

To simplify the notations, let $\theta \equiv \frac{t}{e}$. It decreases in e and increases in t, and represents the relative importance of better access to work and better access to historic city center for households.

Then, if historic city center is within the business area, $q \leq g \leq p$:

$$p = \frac{Ng}{N-2\theta} + \frac{2-N}{2} \text{ and } q = \frac{Ng}{N-2\theta} - \frac{2-N}{2}. \tag{16}$$

If historic city center is to the right to the business area, $q < p \leq g$:

$$p = -\frac{(2-N)N}{4\theta} + \frac{2-N}{2} \text{ and } q = -\frac{(2-N)N}{4\theta} - \frac{2-N}{2}. \tag{17}$$

If historic city center is to the left of business area, $g \leq q < p$:

$$p = \frac{(2-N)N}{4\theta} + \frac{2-N}{2} \text{ and } q = \frac{(2-N)N}{4\theta} - \frac{2-N}{2}. \tag{18}$$

Proposition 3. *If transportation costs are relatively high compare to historic city externality, $\theta \geq \frac{N}{2}$, then the shift of historic center from the geographic city center moves economic center in the opposite direction. However, if increasing g, historic center leaves the business area (or the business area reaches the city boundary), then further increase in g does not affect allocation of firms.*

In contrast, if $\theta < \frac{N}{2}$, then the shift of historic center from the geographic city center moves economic center in the same direction.

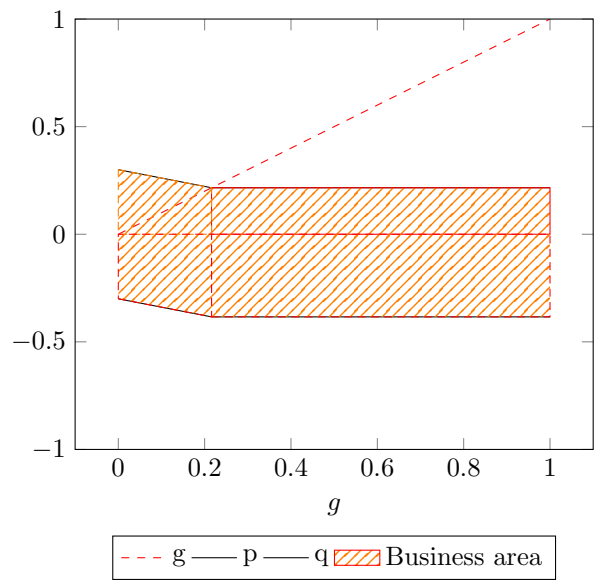

Figure 1. City structure depending on historic city center location, $N = 1.4$, $t = 0.25$ and $e = 0.1$

If historic city center shifts, then an invisible hand will skew the distribution of firms in the city in the opposite direction. So, if one can change the center of beauty externalities or the location of administrative or entertainment center, then he can use it as a tool to affect the spatial organization of economic activity. However, it has certain limitations.

The relation between g and agents allocation is well illustrated in Figure 1. Better access to the historical city center is highly demanded by residents and increases their willingness to pay for their apartments. If $g = 0$, then all geographic, economic, and historic centers coincide. Any deviation of g from the geographical city center encourages residents to "compete" with businesses for center locations. If the historic city center shifts to the "right" side of the city (g increases), then the "right" side of the city becomes more attractive for residents because of the better access to historic city center. It encourage migration of

residents from the "left" to the "right", increasing demand there. As a result, some firms lose the competition for the land in the center area and business area "retreats". Part of residents now commute longer to the job but they agree as now they have better access to historic center. However, the model predicts that when g is outside the business area, allocation of firms does not change as residents stop to compete for land with them as "they got what they want", historic city center. Any further increase in g does not make the right side of the city and do not make it more attractive

3.3 Comparative statics

The assumption that $t > e$ ($\theta > 1$) is quite reasonable as it seems that, on average, we visit workplaces much often than entertainment and historic places. So, costs associated with commuting to work are higher than costs of commuting to beautiful places. However, it seems that we spend much time at home and beautiful surroundings of your place of living also matter a lot. Nowadays, the length of the working day and number of working days per weak decreases, so, people spend more time at home and spend less time on commuting. Many companies start to practice working from home, and remote work become more popular. At the same time, it seems that people start to recognize the importance of architectural heritage, cities authorities understand the role of public spaces in local economies and start to invest more in their development. So, it seems important to consider the effect of t decrease and e increase in this model. Note that $t > e$ implies that $\theta > \frac{N}{2}$ and $\theta > \frac{N(2-N)}{2(2-N+2g)}$. Here we assume that knowledge spillover effects are sufficiently high to satisfy $j > \frac{2t+(2-N)e}{(2-N)V_0}$ for t and e considered. We do not analyse how the type of urban configuration can change with changing in parameters.

Proposition 4. *If residents occupy more land than firms, $N > M$, $g > 0$ and θ decreases then*

1. *if $\theta > \frac{N(2-N)}{2(2-N-2g)}$, then $q < g < p$ and μ decreases;*
2. *if $\frac{N(2-N)}{2(2-N+2g)} < \theta \leq \frac{N(2-N)}{2(2-N-2g)}$, then $q < p \leq g$ and μ decreases;*
3. *if $\frac{M}{2} \leq \theta \leq \frac{N(2-N)}{2(2-N+2g)}$, then all three cases of relative position of g and [q, p] segment are possible in equilibrium;*
4. *if $0 \leq \theta < \frac{M}{2}$, then $q < g < p$ and μ decreases.*

If residents occupy less land than firms, $M \geq N$, $g > 0$ and θ decreases then μ decreases if $\theta \neq \frac{N}{2}$. At $\theta = \frac{N}{2}$ it has a discontinuity.

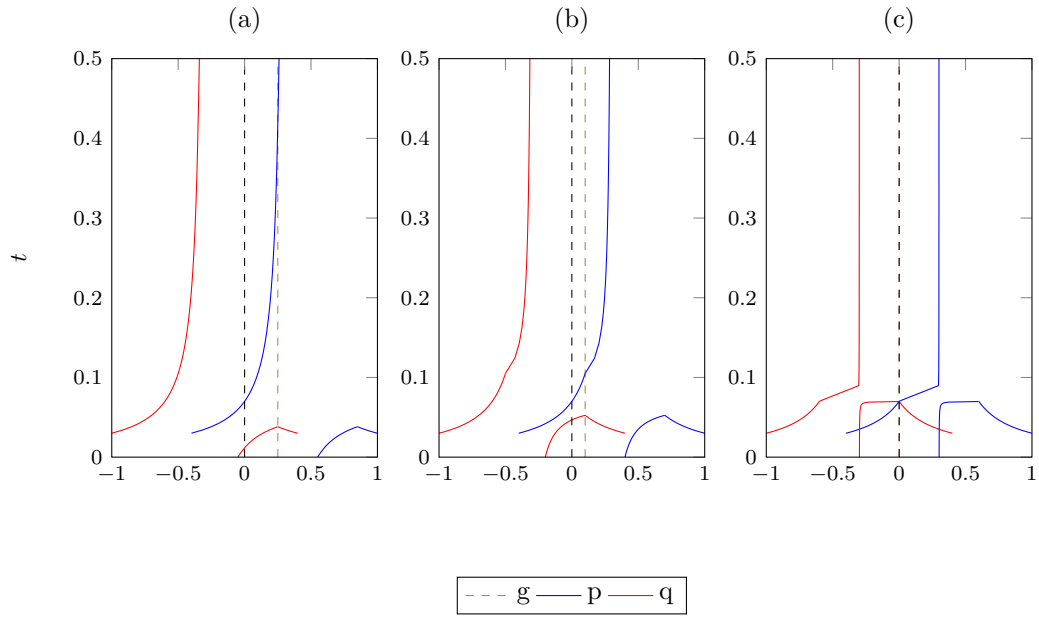

Figure 2. City structure depending on historic city center location, $N > M$, $N = 1.4$, $e = 0.1$ and
(a): $g = 0.25$, (b): $g = 0.1$, (c): $g = 0.001$

If transportation costs are relatively high compare to beauty externalities, then economic city center nearly coincides with geographic city centers. Residents care more about how they will get to the job rather than about beauty or history. People are not ready to have a longer trip to the job but have good access to the historic city center. So, firms that have higher bargaining power win the competition for the city center area.

When commuting costs decrease and beauty externalities increase, people are ready to substitute better access to job by better access to historic city center. So, the economic city center shifts closer to one of the borders.

When θ is too small, economic center comes closer to g. If historic center externalities are extremely high, then firms will choose to locate closer to this center. Rent in this places becomes the highest in the city.

Taking the derivatives of (16), one can also show two important facts. If $q \leq g \leq p$, then decrease in θ increases the effectiveness of g-shifting policies, and, at the same time, higher g increases the effect of decrease in θ.

These relationships are illustrated in Figures 2 and 3. Decreasing transportation costs, we have three possible trajectories of movement of business area that

represented on the graphs.

Note that figures show that an increase in g is associated with a more significant effect of θ-decrease on firms allocation, so, it is possible to affect the distribution of agents even by a little commuting costs reduction. If g is high, then even at high t it is possible to affect the distribution of people. Lower t increases the effectiveness of shifting g. Supporting Proposition 1, it is also shown that higher g is associated with higher shift of businesses for each t. So, if city planner will want to use the described mechanisms, he should be aware that transportation costs are low, while historic city center value is socially recognizable or new entertainment center is highly demanded.

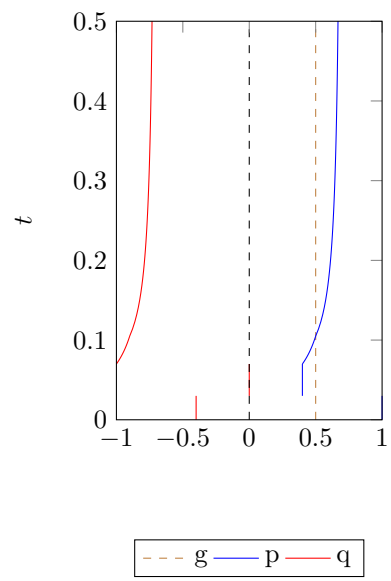

Figure 3. City structure depending on historic city center location, $M > N$, $N = 0.6$, $e = 0.1$ and $g = 0.5$

Note that there is an area where multiple equilibria are possible and discontinuity in economic center allocation trajectory, so, the model is characterized by bifurcation dynamic behavior for some t. Gradually decreasing t from high values to ones bellow $\frac{Me}{2}$ level, results in jump of economic activity from one side of the city to another. However, if then t will gradually increase back, it may happen so that business will not return to its initial positions.

4 Optimal historic city center

In order to study how historic and economic city centers should be located, we analyse what is the best place for historic center. It can be assumed that we consider a problem of the planner that should find the most optimal place for park or public offices. Analysing the effect of historic city location we look at its impact on total welfare in the economy, which is defined as a net income spent on consumption by residents and the overall land rent.

$$TW = \int_{-1}^{1} R(x)\,dx + NI_0. \tag{19}$$

Theorem 1. *Social welfare is maximized at* $g = \pm\frac{M}{2}$ *if* $\theta \geq \frac{N}{2}$*, and at* $g = 0$ *if* $\theta < \frac{N}{2}$*.*

It is a very powerful theorem as it shows that usually ($t > e$) it is socially desirable to separate historic and economic city centers. Moreover, it should be located at residential area and the exact location does not depend on the strength of externality, transportation costs, and knowledge spillover effects. This result is obtained just by maximizing rent in the city meaning that it generates the most efficient allocation of agents in space. This result is explained not by ideological or cultural reasons as the threat of losing historical heritage.

5 Long run spatial equilibrium

Consider the transition from short-run equilibrium already described to the long run one. Note that, in general, there is a jump discontinuity of rent on one of the borders of the city. Near this border, people have to pay much higher rents if they are within the city compare to rent outside. So, people will encounter locational arbitrage possibilities and have incentives to relocate a little bit beyond the city boundary. It will not significantly raise the commuting costs but will allow to economize on rent payments. So, such individual reallocation of people will move the whole city in one direction, which is associated with rent discontinuity. City will move until rent will equalize on both boundaries, $R(-1) = R(z) = 1$.

Note that if $R_c(-1) - R_c(1) > 0$, then city moves to the left. If g is fixed, then boundaries coordinates decreases. Alternatively, if we want to save the coordinate system of the system on the [-1, 1] domain, then g is interpreted as a relative position of historic city center, then shift of the city to the left means increase in g. Similarly, if $R_c(-1) - R_c(1) < 0$, then city moves to the right and g decreases.

Theorem 2. *In case of free mobility, for given N, t and $e < j$, there exists only three values of g that can sustain the long run competitive spatial monocentric equilibrium: $g = \pm\frac{M}{2}$ and $g = 0$.*

If $\theta > \frac{N(2-N)}{2(2-N+2g)}$, two equilibria with $g = \pm\frac{M}{2}$ are stable, while equilibrium with $g = 0$ is not stable.

If commuting costs are low, $\theta < \frac{N(2-N)}{2(-2+N+2g)}$, only $g = 0$ can sustain the long run competitive spatial monocentric equilibrium.

If $\frac{N(2-N)}{2(-2+N+2g)} \leq \theta \leq \frac{N(2-N)}{2(2-N+2g)}$, there are stable equilibria with $g = \pm\frac{M}{2}$ and unstable with $g = 0$ if $p < g$, and there is stable equilibrium with $g = 0$ if $p > g$.

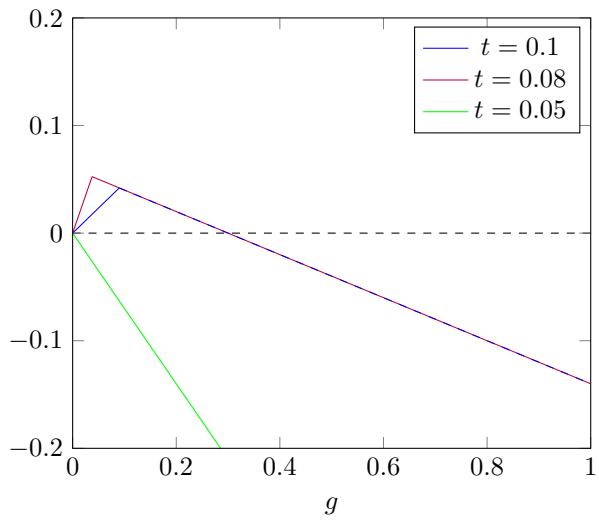

Figure 4. Border rent discontinuity, $R(-1) - R(1)$, $N = 1.4$ and $e = 0.1$

It means that even very little deviation of g from geographical city center, in the long run, should lead to the further shift of the whole city increasing $|g|$ to $|\frac{M}{2}|$ level. So, according to this model in order to completely separate the economic and historic city centers, it is sufficient to make a little shift in g and then market forces, human incentives, without any introduction of taxation or land regulation will do this job.

In reality, people and firms are not fully mobile, changing the location may associate with some costs. Residential and industrial buildings are not perfect substitutes. And what is the most important, the organization and location of the city are determined by preexisting infrastructure. Our model ignores density issues assuming uniform land distribution, while in reality, it is hard to "shift"

the density of population as it is not easy to move skyscrapers. So, it is not so easy to move the city and relocate the business within the city, and predictions of the model will not necessarily hold. Nevertheless, this work provides impressive insights on city organization patterns, incentives of people within the city, and their choices of living. It provides a mechanism that explains how markets force cities to reorganization, separating economic and historic city centers.

Assume that agents are not free mobile and there are some fixed costs of leaving the city as infrastructure and buildings are needed to be built outside the city borders. So, the cost of leaving the city is c. Then, agents will decide to leave the city if border rent discontinuity, $|R_c(-1) - R_c(1)|$, should be higher then c. Thus, little change in g may not shift the whole city. Lower t and higher e increases $|R_c(-1) - R_c(1)|$ meaning that lower g is sufficient to launch shift of the city (see Figure 4). So, the fact cities may not react to g changes means that θ is not sufficiently low.

6 Beauty externalities at workplace

As it was already mentioned, people may also benefit from the fact that their workplace is near the historic city center. They can be happier if they will work in beautiful place or have an opportunity to visit gallery or museum after the work. So, it is relevant to consider the effect of presence of beauty externalities at workplace (or commuting costs from workplace to historic city center). We relax assumption that $v = 0$, considering (8). Therefore, resident rent function (11) is the same, while (12) can be rewritten as

$$R_f(z) = \begin{cases} V_0Q(z) - \frac{N}{M}(R_c(1) + t(1-z) + e(1-g) + v|g-z| + I_0), & \text{if } z \geq \hat{\mu}, \\ V_0Q(z) - \frac{N}{M}(R_c(-1) + t(1+z) + e(1+g) + v|g-z| + I_0), & \text{if } z < \hat{\mu}. \end{cases}$$

Taking $v > 0$, we model the fact that firms want to be closer to the historic city center. In our model it happens because it allows them to pay less for the workers.

Border rent discontinuity is

$$R_c(-1) - R_c(1) = \frac{2e - N(e+v)}{2-N}(|g-q| - |g-p|) - 2eg, \tag{20}$$

So, the economic city center location is

$$\mu = \begin{cases} \frac{((e+v)N-2e)Ng}{(e+v)N^2-2(e+t)N+4t}, & \text{if } q \leq g \leq p, \\ \frac{N^2e+N^2v-2Ne}{4t}, & \text{if } q < p \leq g, \\ -\frac{N^2e+N^2v-2Ne}{4t}, & \text{if } g \leq q < p. \end{cases} \tag{21}$$

The main difference that v bring to the analysis compare to previous sections is that now the measure of relative importance of access to work and access to historic city center is $\theta = \frac{t}{e - \frac{N}{2-N}v}$. θ increases in v and decreases in e. It increases in t if $v < \frac{(2-N)e}{N}$, and decreases otherwise.

Taking the derivatives of (21), it can be shown that μ decreases in e if $g \geq q$, and increases in e otherwise, repeating results from previous sections. At the same time, there is a positive relation between v and μ for $g \geq q$, and negative otherwise.

So, adding the commuting between workplace and historic center in the analysis lead to the decrease of the effect of beauty externalities on the separation of city centers. The main driver of this process was the increasing attractiveness of places near historic city center. If now we allow people to visit it from the workplace, it becomes less reasonable to live near historic city center but find such job place.

The analysis of the effect of transportation costs on business activities become more complex. If $v \leq \frac{(2-N)e}{N}$, i.e. $\theta > 0$, then increase in transportation costs lead to increase in μ, shifting the whole business area to the right. If $v \geq \frac{(2-N)e}{N}$, the effect of transportation costs rise will be the opposite.

If the importance of beauty at workplace is not very high, $v \leq \frac{(2-N)e}{N}$, i.e. $\theta \geq 0$, then all the mechanisms in the model will be the same as in previous sections. However, their strength will be weaker and degree of separation of economic and historic centers will be lower. So, non-zero long-run spatial equilibrium will be at $g = \pm(\frac{M}{2} - \frac{Nv}{2e})$.

If the importance of beauty at workplace is relatively high, firms will be willing more to locate near the historic city center increasing demand there. If $v \geq \frac{(2-N)e}{N}$, i.e. $\theta \leq 0$, then g will be inside the business area. Any shifting of historic city center will shift the whole business area in the same direction. The only possible long-run spatial equilibrium will be $g = 0$.

So, the increased importance of the connection between workplace and historic city center can significantly affects our conclusions on city centers separability.

7 Home-monocentric urban configuration

In this section we look at situations when residents are located near the center of the city and are surrounded by firms. Home-monocentric city configuration can take place only if $j < \frac{-2t+(2-N)e}{(2-N)V_0}$, which implies $\theta < \frac{2-N}{2}$.

In this case, $\hat{\mu} = -\frac{(R_f(-1)-R_f(1))(2-N)}{2tN}$, and wages are higher for firms located near the city borders. Border rent discontinuity is

$$R_c(-1) - R_c(1) = e(|g - p_h| - |g - q_h|) - jV_0(|p_h| - |q_h|), \tag{22}$$

If we define, the residential center location as $\nu = \frac{p_h+q_h}{2}$, then

$$\nu = \begin{cases} \frac{(2-N)^2 ge}{(2-N)^2(e-jV_0)+2Nt}, & \text{if } q_h \leq g \leq p_h, \\ \frac{(2-N)^2(jV_0+e)}{4t}, & \text{if } q_h < p_h \leq g, \\ -\frac{(2-N)^2(jV_0+e)}{4t}, & \text{if } g \leq q_h < p_h. \end{cases} \tag{23}$$

If $q \leq g \leq p$ increase in g leads to the increase in ν. It means that the shift of historic center from the geographic city center moves economic center in the same direction. However, if increasing g, historic center leaves the residential area, then further increase in g does not affect allocation of residents. The idea is that households prefer to be near the historic centers and are ready to migrate following its movements.

Increase in θ lead to decrease in ν for $g \geq q$, meaning that decrease in transpiration costs and increased beauty externalities increases the attractiveness of places near historic city center. Residents are less worried about access to workplaces and start to migrate closer to such areas. Taking relevant derivatives of (23), one can derive that for lower θ agents react stronger on any changes in g. For higher g agents react stronger on any changes in θ.

All these conclusions are illustrated on the Figure .

Moreover, the only possible long-run spatial monocentric urban configuration exists at $g = 0$, which is at the same time maximizes social welfare. So, if beauty externalities become too high and home-monocentric urban configuration emerge, then one should expect symmetric city structure around the historic city center. Despite the fact that formally in this model it means that μ and g coincides, firms will locate at the boundaries of the city and should not threaten the existence of historic center. So, historic city center and business environment will be separated. This result is self-driven and the most optimal for the society.

8 Conclusion

In this work, we develop a model of the city with two possible sources of externalities. First, we show that monocentric city configuration can emerge only

if knowledge spillovers effects can overcome the effect of historic city center externalities. In other words, agglomeration forces that unites firms should be stronger than ones that attract residents.

The most important result of the paper is that there exists a social demand for separation of "historic" and "economic" city centers if commuting costs are relatively high or if beauty externalities are not high. Moreover, in the long run market forces can bring city to this outcome without any land regulation and taxation. City centers separability will shift the whole agglomeration density. It means that in theory Brussalization problem can be easily solved: the social planner should slightly change the allocation of cultural and historical clusters in the city, then it will awaken market forces that will separate economic and historic centers increasing social welfare. The main idea is that social planner should generate border rate discontinuity. In practice, he can temporally subsidise the rent outside the city at one side, build necessary infrastructure decreasing deterrent forces and increasing human incentives to reallocate outside the city.

So, this work provides evidence that there may occur a conflict between "historic" and "economic" city centers and it can be optimal to separate them. Lower commuting costs and higher beauty externalities make "historic center reallocation policy" more effective. It allows to move business activities away from the historic center, so, economic and historic city centers move in opposite directions. However, too high level of beauty externalities and too low commuting costs implies the higher chances to have cultural and economic centers closer to the geographical city center. If commuting costs are too low, then it is better to not separate the centers.

Monocentric city configuration can emerge only if knowledge spillovers effects can overcome the effect of historic city center externalities. Note that knowledge spillovers determine the type of city configuration, while commuting costs shape the allocation of agents and city centers. Changes in the strength of agglomeration forces between firms do not change anything except chances to have monocentric city configuration.

In future works, we can study other possible city configurations. Importantly, this work shows that there exists a demand for distinguished centers in the city. Thus, the proposed model probably can explain the nature of central business district blurring and offer a mechanism for the genesis of polycentric cities. Reformulation of agglomeration forces between firms will allow us to get the model of polycentric urban configurations. It seems reasonable to consider better access to labor as the main force that brings firms to the center.

It is reasonable to study the transition to the long-run equilibrium of growing city more accurately. We want to consider not uniformly land distribution, but allow developers or landlord to change it building durable tall houses. The

dynamic of changes in the density of population and "moving" the city may also explain the existence of polycentric cities.

We can introduce heterogeneity in production and productivity in the city. More productive firms and residents may allocate in the center or near historic center, while less productive agents will locate near the boundaries of the city forming subcenters that may generate polycentric patterns.

Also, we can consider heterogeneous types of residents with different e and t parameters to capture differences between financial and tourist cities, which have different types of people.

In this work, we want to understand whether it is possible to distinguish economic and historic city centers. In our analysis, we assume that historic city center is exogenous, however, in reality, it is not the case. As economic city center is a place where rents seems to be the highest and people spend much time, it seems that investments in infrastructure and beauty are higher there. So, it is highly important to understand how beauty is "produced" and which agents will be able to maintain it. Also, it is important to understand that firms may also be interested in situating near the beautiful places, as it can motivate workers increasing their productivity or increase customers' flows. Thus, if business play a significant role in beauty creation, it also should be taken into account in the analysis of the allocation of centers.

9 Appendix

Proposition 5. *A competitive spatial job-monocentric equilibrium exists if knowledge spillover effects are relatively high (compare to historic city center externalities effects),* $j > \frac{2t+(2-N)e}{(2-N)V_0}$.

A competitive spatial home-monocentric equilibrium exists if knowledge spillover effects are relatively low (compare to historic city center externalities effects), $j < \frac{-2t+(2-N)e}{(2-N)V_0}$.

Proof. First, suppose that competitive spatial job-monocentric equilibrium exists. All firms locate within the interval $[q, p]$.

A competitive spatial job-monocentric equilibrium exists if $|\frac{dR_f(x)}{dx}| > |\frac{dR_c(x)}{dx}|$ at locations p and q. Taking the derivatives of (11) and (12), one can derive that $|\frac{dR_f(x)}{dx}| = V_0 j - \frac{N}{2-N}t$ and $\frac{dR_c(x)}{dx}(x = q) = e + t$, $\frac{dR_c(x)}{dx}(x = p) = \pm e - t$. So, equilibrium simply exists when $V_0 j - \frac{N}{2-N}t > e + t$. It can be rewritten as $j > \frac{2t+(2-N)e}{(2-N)V_0}$ Note that $R_f(x)$ and $R_c(x)$ are equalized at q and p by I_0 term.

Now, suppose that competitive spatial home-monocentric equilibrium exists.

All residents live within the interval $[q_h, p_h]$.

Consider a resident that live at location $x = \hat{\mu}$ which is indifferent whether to go to work to the right or left side of the city. The wages in the economy should allow him to work at any workplace which located to the right or to the left of $\hat{\mu}$. Then, manipulating (8) one can get

$$w(z) = \begin{cases} R_c(\hat{\mu}) + t(z - \hat{\mu}) + e|g - \hat{\mu}| + I_0, \text{ if } z \geq \hat{\mu}, \\ R_c(\hat{\mu}) + t(\hat{\mu} - z) + e|g - \hat{\mu}| + I_0, \text{ if } z < \hat{\mu}. \end{cases} \tag{24}$$

Then, substituting (24) into (8) yields:

$$R_c(z) = \begin{cases} R_c(\hat{\mu}) + t(z - \hat{\mu}) + e|g - \hat{\mu}| - e|g - z|, & \text{if } z \geq \hat{\mu}, \\ R_c(\hat{\mu}) + t(\hat{\mu} - z) + e|g - \hat{\mu}| - e|g - z|, & \text{if } z < \hat{\mu}. \end{cases} \tag{25}$$

Combining (7) with (24),

$$R_f(z) = \begin{cases} V_0 Q(z) - \frac{N}{M}(R_c(\hat{\mu}) + t(z - \hat{\mu}) + e|g - \hat{\mu}| + I_0), & \text{if } z \geq \hat{\mu}, \\ V_0 Q(z) - \frac{N}{M}(R_c(\hat{\mu}) + t(\hat{\mu} - z) + e|g - \hat{\mu}| + I_0), & \text{if } z < \hat{\mu}. \end{cases} \tag{26}$$

A competitive spatial home-monocentric equilibrium exists if $|\frac{dR_c(x)}{dx}| > |\frac{dR_f(x)}{dx}|$ at locations p_h and q_h. Taking the derivatives of (25) and (26), one can derive that the relevant condition is $j < \frac{-2t+(2-N)e}{(2-N)V_0}$.

Note that $\frac{-2t+(2-N)e}{(2-N)V_0} < \frac{2t+(2-N)e}{(2-N)V_0}$. □

Proposition 6. *In the model proposed, polycentric urban configurations cannot emerge.*

Proof. Consider case of job-polycentric urban configuration, situation when two (or more) clusters of firms exist, which are surrounded by residents. Consider an interval $[\hat{\mu}_1, \hat{\mu}_3]$ with two business areas $[q_1, p_1]$ and $[q_2, p_2]$ $(q_1 < p_1 < q_2 < p_2)$. Then, there exists $\hat{\mu}_2$ $(p_1 < \hat{\mu}_2 < q_2)$ such that all residents on the interval $[\hat{\mu}_1, \hat{\mu}_2]$ work at $[q_1, p_1]$ area, while all residents in $[\hat{\mu}_2, \hat{\mu}_3]$ work at $[q_2, p_2]$. Then wages are

$$w(z) = \begin{cases} R_c(\hat{\mu}_1) + t(z - \hat{\mu}_1) + e|g - \hat{\mu}_1| + I_0, \text{ if } \hat{\mu}_1 \leq z \leq \hat{\mu}_{11}, \\ R_c(\hat{\mu}_2) + t(\hat{\mu}_2 - z) + e|g - \hat{\mu}_2| + I_0, \text{ if } \hat{\mu}_{11} \leq z \leq \hat{\mu}_2, \\ R_c(\hat{\mu}_2) + t(z - \hat{\mu}_2) + e|g - \hat{\mu}_2| + I_0, \text{ if } \hat{\mu}_2 \leq z \leq \hat{\mu}_{12}, \\ R_c(\hat{\mu}_3) + t(\hat{\mu}_3 - z) + e|g - \hat{\mu}_3| + I_0, \text{ if } \hat{\mu}_{12} \leq z \leq \hat{\mu}_3. \end{cases} \tag{27}$$

$$R_c(z) = \begin{cases} R_c(\hat{\mu}_1) + t(z - \hat{\mu}_1) + e|g - \hat{\mu}_1| - e|g - z|, \text{ if } \hat{\mu}_1 \leq z \leq \hat{\mu}_{11}, \\ R_c(\hat{\mu}_2) + t(\hat{\mu}_2 - z) + e|g - \hat{\mu}_2| - e|g - z|, \text{ if } \hat{\mu}_{11} \leq z \leq \hat{\mu}_2, \\ R_c(\hat{\mu}_2) + t(z - \hat{\mu}_2) + e|g - \hat{\mu}_2| - e|g - z|, \text{ if } \hat{\mu}_2 \leq z \leq \hat{\mu}_{12}, \\ R_c(\hat{\mu}_3) + t(\hat{\mu}_3 - z) + e|g - \hat{\mu}_3| - e|g - z|, \text{ if } \hat{\mu}_{12} \leq z \leq \hat{\mu}_3. \end{cases} \tag{28}$$

$$R_f(z) = \begin{cases} V_0Q(z) - \frac{N}{M}(R_c(\hat{\mu}_1) + t(z - \hat{\mu}_1) + e|g - \hat{\mu}_1| + I_0), & \text{if } \hat{\mu}_1 \leq z \leq \hat{\mu}_{11}, \\ V_0Q(z) - \frac{N}{M}(R_c(\hat{\mu}_2) + t(\hat{\mu}_2 - z) + e|g - \hat{\mu}_2| + I_0), & \text{if } \hat{\mu}_{11} \leq z \leq \hat{\mu}_2, \\ V_0Q(z) - \frac{N}{M}(R_c(\hat{\mu}_2) + t(z - \hat{\mu}_2) + e|g - \hat{\mu}_2| + I_0), & \text{if } \hat{\mu}_2 \leq z \leq \hat{\mu}_{12}, \\ V_0Q(z) - \frac{N}{M}(R_c(\hat{\mu}_3) + t(\hat{\mu}_3 - z) + e|g - \hat{\mu}_3| + I_0), & \text{if } \hat{\mu}_{12} \leq z \leq \hat{\mu}_3. \end{cases} \tag{29}$$

Then one take to analyse the derivatives of rent functions at q_1, p_1, q_2, and p_2.

At $z = q_1$ we have $\frac{dR_c(x)}{dx} = t + s_{q_{11}}e < \frac{dR_f(x)}{dx} = s_{q_{21}}V_0j - \frac{N}{M}t$.
At $z = p_1$ we have $\frac{dR_c(x)}{dx} = -t + s_{p_{11}}e > \frac{dR_f(x)}{dx} = s_{p_{21}}V_0j + \frac{N}{M}t$.
At $z = q_2$ we have $\frac{dR_c(x)}{dx} = t + s_{q_{12}}e < \frac{dR_f(x)}{dx} = s_{q_{22}}V_0j - \frac{N}{M}t$.
At $z = p_2$ we have $\frac{dR_c(x)}{dx} = -t + s_{p_{12}}e > \frac{dR_f(x)}{dx} = s_{p_{22}}V_0j + \frac{N}{M}t$.

$s_{q_{ij}} = \pm 1$ and $s_{p_{ij}} = \pm 1$, such that $s_{q_{11}} \leq s_{p_{11}} \leq s_{q_{12}} \leq s_{p_{12}}$ and $s_{q_{21}} \leq s_{p_{21}} \leq s_{q_{22}} \leq s_{p_{22}}$.

Suppose that $s_{q_{11}} = 1$, then $s_{q_{21}}$ should be equal to 1.
Then, despite the sign of $s_{p_{11}}$, $s_{p_{21}}$ can be only -1 to satisfy the condition at $z = q_1$.
Then, $s_{q_{22}} = s_{p_{22}} = -1$.
Then, condition at $z = q_2$ can hold only if $s_{q_{12}} = -1$ and $e > t$.
Then, taking sums of left-hand sides and right-hand sides of inequalities at $z = q_1$ and $z = q_2$, one will get that $t < 0$, which is not true.

Suppose that $s_{q_{11}} = -1$, then $s_{p_{11}} = s_{q_{12}} = s_{p_{12}} = -1$.
Then, if $s_{q_{21}} = 1$, then condition at $z = p_1$ can hold only if $s_{p_{21}} = -1$.
Then, $s_{q_{22}} = s_{p_{22}} = -1$.
Then, inequalities at $z = p_1$ and $z = q_2$ can hold only if $t < 0$, which is not true.

Suppose that $s_{q_{11}} = -1$, then $s_{p_{11}} = s_{q_{12}} = s_{p_{12}} = -1$.
Then, if $s_{q_{21}} = -1$, then $s_{p_{21}} = s_{q_{22}} = s_{p_{22}} = -1$.
Then, inequalities at $z = p_1$ and $z = q_1$ can hold only if $t < 0$, which is not true.

So, it is impossible to have urban configuration with two (or more) clusters of firms. Similarly, can be proved that urban configuration with two (or more) clusters of residents is also not possible.

□

Proposition 7. *If transportation costs are relatively high compare to historic city externality, $\theta \geq \frac{N}{2}$, then the shift of historic center from the geographic city center moves economic center in the opposite direction. However, if increasing g, historic center leaves the business area (or the business area reaches the city boundary), then further increase in g does not affect allocation of firms.*

In contrast, if $\theta < \frac{N}{2}$, then the shift of historic center from the geographic city center moves economic center in the same direction.

Proof. If historic city center is within the business area, $q \leq g \leq p$, then economic city center is $\mu = \frac{Neg}{2(Ne-2t)} = \frac{Ng}{N-2\theta}$. Note that if $g = 0$, then city is symmetric and $q < g < p$. If g increases, then μ decreases for $\theta = \frac{t}{e} \geq \frac{N}{2}$. When $g = \frac{(2-N)(2t-eN)}{4t} = -\frac{(2-N)N}{4\theta} + \frac{2-N}{2}$, it coincides with the right business area boundary, and further increase in g is interpreted as that historic center leaves this area.

When $g > \frac{(2-N)(2t-eN)}{4t}$, then $q < p < g$ and and boundaries are determined by (17). As we can see, they do not depend on g.

If $\theta = \frac{t}{e} < \frac{N}{2}$, then μ increases until business area reaches the right boundary of the city. □

Proposition 8. *If residents occupy more land than firms, $N > M$, $g > 0$ and θ decreases then*

1. *if $\theta > \frac{N(2-N)}{2(2-N-2g)}$, then $q < g < p$ and μ decreases;*
2. *if $\frac{N(2-N)}{2(2-N+2g)} < \theta \leq \frac{N(2-N)}{2(2-N-2g)}$, then $q < p \leq g$ and μ decreases;*
3. *if $\frac{M}{2} \leq \theta \leq \frac{N(2-N)}{2(2-N+2g)}$, then all three cases of relative position of g and [q, p] segment are possible in equilibrium;*
4. *if $0 \leq \theta < \frac{M}{2}$, then $q < g < p$ and μ decreases.*

If residents occupy less land than firms, $M \geq N$, $g > 0$ and θ decreases then μ decreases if $\theta \neq \frac{N}{2}$. At $\theta = \frac{N}{2}$ it has a discontinuity.

Proof. Analyzing and manipulating (16), (17) and (18), one can find the conditions when each of the cases is possible. Taking into account that $-1 \leq q$ and $p \leq 1$ and comparing p and q with g, one can derive conditions on θ.

Considering all the cases, economic center location is $\mu = \frac{Ng}{N-2\theta}$ or $\mu = -\frac{(2-N)N}{4\theta}$ if $q < g$. Both functions are increasing in θ. If $g \leq q$, then $\mu = -\frac{(2-N)N}{4\theta}$. □

Theorem 3. *Social welfare is maximized at $g = \pm\frac{M}{2}$ if $\theta \geq \frac{N}{2}$, and at $g = 0$ if $\theta < \frac{N}{2}$.*

Proof. For each trajectory specified in Proposition 3, substituting (14), (16), (17) and (18) into (19), it is possible to calculate $TW(N, g, e, t)$. Then differentiation with respect to g allow us to find location of g that maximizes it.

Consider all possible relations between p, g and q. If $q < g < p$, then

$$TW = \frac{e(-V_0 j + e)N^3 + (-4e^2 + ((4j-4)V_0 - 4t)e + 2V_0 jt)N^2}{4Ne - 8t} + \frac{(-4e^2 g^2 + ((-4j+8)V_0 + 8t)e - 8t((j-1)V_0 - 0.5t))N + 8tV_0(j-2)}{4Ne - 8t}$$

$$\frac{dTW(g)}{dg} = -\frac{8e^2 Ng}{4Ne - 8t}$$

$$\frac{d^2TW(g)}{dg^2} = -\frac{8e^2 N}{4Ne - 8t}$$

So, $g = 0$ is the local maximum for $t < \frac{Ne}{2}$, and minimum otherwise. If $q < p \leq g$, then

$$TW = \frac{-4Nt^2 + (-2N^2 jV_0 + (-8ge + (8(j-1))V_0)N}{8t} + \frac{-8(g-1)^2 e - (8(j-2))V_0)t + Ne^2(N-2)^2}{8t}$$

$$\frac{dTW(g)}{dg} = -eN - 2e(g-1)$$

$$\frac{d^2TW(g)}{dg^2} = -2e$$

Then, $g = \frac{2-N}{2}$ is the local maximum.

If $g \leq q < p$, then

$$TW = \frac{-4Nt^2 + (-2N^2 jV_0 + (8ge + (8(j-1))V_0)N}{8t} + \frac{-8(g+1)^2 e - (8(j-2))V_0)t + Ne^2(N-2)^2}{8t}$$

$$\frac{dTW(g)}{dg} = eN - 2e(g+1)$$

$$\frac{d^2TW(g)}{dg^2} = -2e$$

Then, $g = -\frac{2-N}{2}$ is the local maximum, however, as $g \geq 0$ it is not a solution and $g = q$ is local maximum. Note that then $TW(g = 0|q < g < p) \geq TW(g = q|g \leq q < p)$.

Comparing TW for different trajectories, $t \geq \frac{Ne}{2}$, then $TW(g = \frac{2-N}{2}|q < p \leq g) \geq TW(g = 0|q < g < p)$. $g = 0$ maximizes TW if $t < \frac{Ne}{2}$ $\square$

Theorem 4. *In case of free mobility, for given N, t and $e < j$, there exists only three values of g that can sustain the long run competitive spatial monocentric equilibrium: $g = \pm\frac{M}{2}$ and $g = 0$.*

If $\theta > \frac{N(2-N)}{2(2-N+2g)}$, two equilibria with $g = \pm\frac{M}{2}$ are stable, while equilibrium with $g = 0$ is not stable.

If commuting costs are low, $\theta < \frac{N(2-N)}{2(-2+N+2g)}$, only $g = 0$ can sustain the long run competitive spatial monocentric equilibrium.

If $\frac{N(2-N)}{2(-2+N+2g)} \leq \theta \leq \frac{N(2-N)}{2(2-N+2g)}$, there are stable equilibria with $g = \pm\frac{M}{2}$ and unstable with $g = 0$ if $p < g$, and there is stable equilibrium with $g = 0$ if $p > g$.

Proof. Considering (14), $R_c(-1) - R_c(1) = 0$ at $g = 0$ and $g = \frac{2-N}{2}$. Substituting (16), (17) and (18), one can derive that

$$R_c(-1) - R_c(1) = \frac{2e^2Ng}{2t - eN} \text{ when } q < g < p,$$

$$R_c(-1) - R_c(1) = e(2 - N - 2g) \text{ when } q < p \leq g,$$

$$R_c(-1) - R_c(1) = e(-2 + N - 2g) \text{ when } g \leq q < p.$$

Note that $R_c(-1) - R_c(1)$ reaches positive values at $g < \frac{2-N}{2}$ and negative at $g > \frac{2-N}{2}$ if $t \geq \frac{eN(2-N)}{2(2-N+2g)}$. If $t \leq \frac{e(2-N)}{2}$, then $R_c(-1) - R_c(1)$ is negative for any g. For t that multiple equilibria are possible, $R_c(-1) - R_c(1)$ depends on which equilibrium is considered. □

In case of home-monocentric urban configuration, masses of agents can be rewritten as $M = 2 - p_h + q_h$ and $N = p_h - q_h$.

Consider $R(z)$ at the boundaries of the city, substituting $z = 1$ and $z = -1$ into (**??**). Then, taking the differences, one can get $R_f(-1) - R_f(-1) = -\frac{2N\hat{\mu}t}{2-N}$, therefore, $\hat{\mu} = -\frac{(2-N)(R_f(-1)-R_f(-1))}{2tN}$. Then, remind that all "left" firms hire "left" workers and all "right" firms hire "right" workers. So, $\frac{1+q_h}{\hat{\mu}-q_h} = \frac{1-p_h}{p_h-\hat{\mu}} = \frac{M}{N}$. Therefore,

$$p_f = -\frac{(2-N)^2(R_f(-1) - R_f(-1))}{4Nt} + \frac{N}{2}$$

$$q_f = -\frac{(2-N)^2(R_f(-1) - R_f(-1))}{4Nt} - \frac{N}{2}$$

Note that $\mu = 0$, then $Q(p_h) = 1 - j|p_h|$ and $Q(q_h) = 1 - j|q_h|$. $R(z)$ is continuous, so, $R_c(p_h) = R_f(p_h)$ and $R_c(q_h) = R_f(q_h)$. Then, by manipulating these

expressions, we get (**??**):

$$R_c(-1) - R_c(1) = e(|g - p_h| - |g - q_h|) - jV_0(|p_h| - |q_h|).$$

Then, considering all cases for module terms, one can derive (23):

$$\nu = \begin{cases} \frac{(2-N)^2 ge}{(2-N)^2(e-jV_0)+2Nt}, & \text{if } q_h \leq g \leq p_h, \\ \frac{(2-N)^2(jV_0+e)}{4t}, & \text{if } q_h < p_h \leq g, \\ -\frac{(2-N)^2(jV_0+e)}{4t}, & \text{if } g \leq q_h < p_h. \end{cases}$$

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